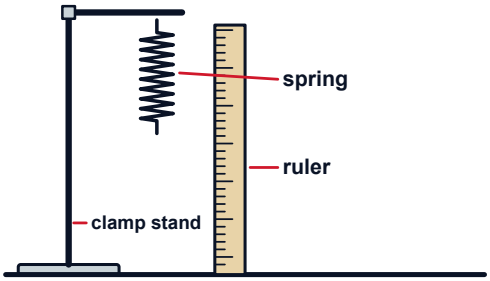
DO NOW · REQUIRED PRACTICAL

Force and extension of a spring

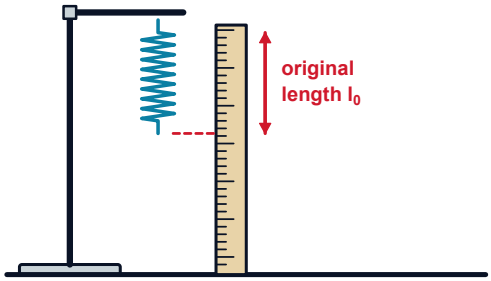
Forces
Paper 2 · Combined and Separate
6 steps

A student investigates how the force on a spring affects its extension.

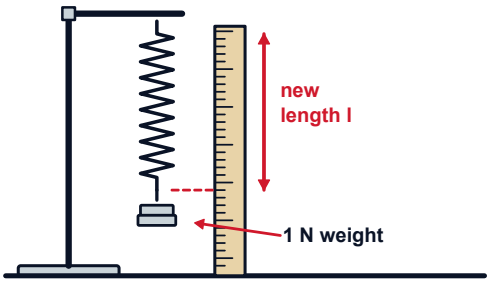
1 Hang the spring from a clamp stand next to a ruler.



2 Measure the original length of the spring.



3 Hang a 1 N weight on the spring and measure its new length.



4 Calculate the extension: new length − original length.

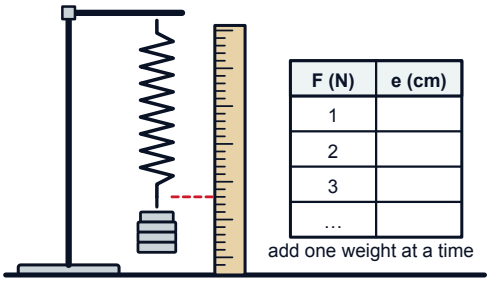
$$e = l - l_0$$

extension = new length − original length



no weight | with weight

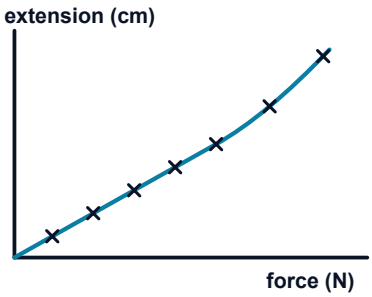
5 Add more weights one at a time, finding the extension each time.



F (N)	e (cm)
1	
2	
3	
…	

add one weight at a time

6 Plot a graph of extension against force.



extension against force

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